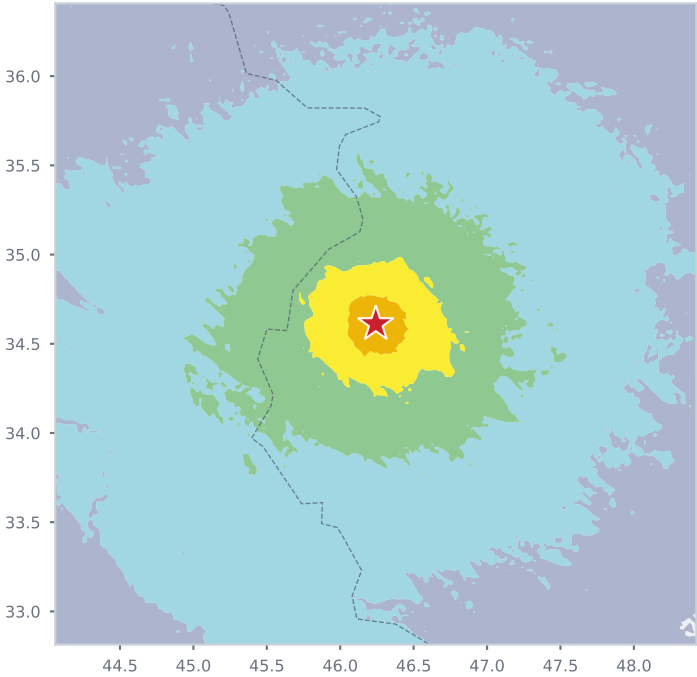
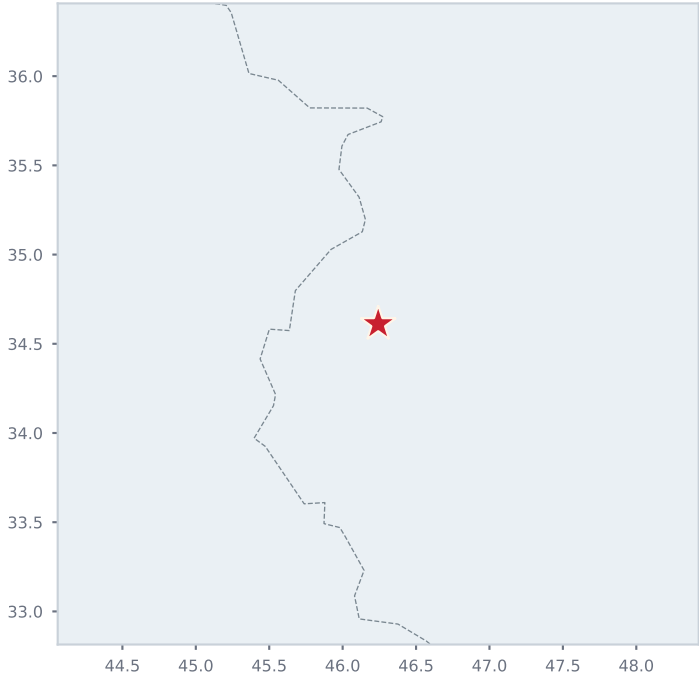


Shaking intensity (IMS-25)



Expected building damage grade



IMS-25 intensity



Expected damage grade (IMS-25)



Damage alert

ORANGE

Widespread damage; many buildings unusable

Provisional

Night occupancy

Buildings heavily damaged (DG3 or worse)

480 to 4.8 thousand

Central estimate 1.4 thousand. The range is the 5th to 95th percentile.

Buildings in the shaken area

4.4 million

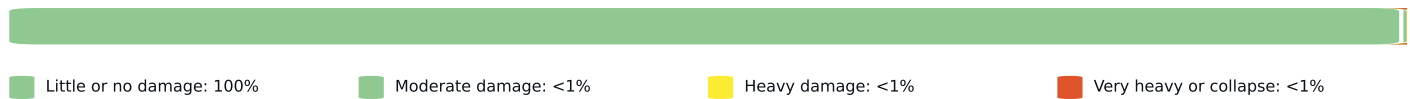
21% of them inferred from built-up area, not observed footprints.

People by shaking level



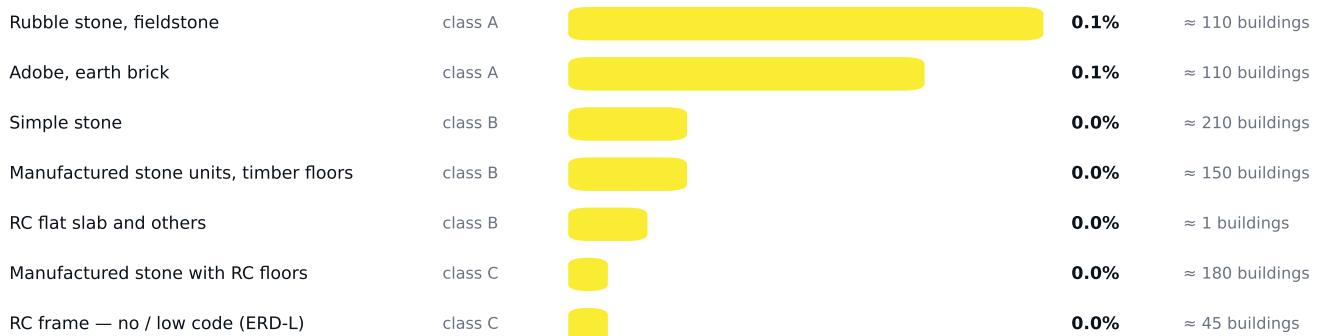
Building exposure modelled for: Iraq. Shaded areas without exposure data are shown for shaking only.

Share of buildings in the mapped area by expected damage

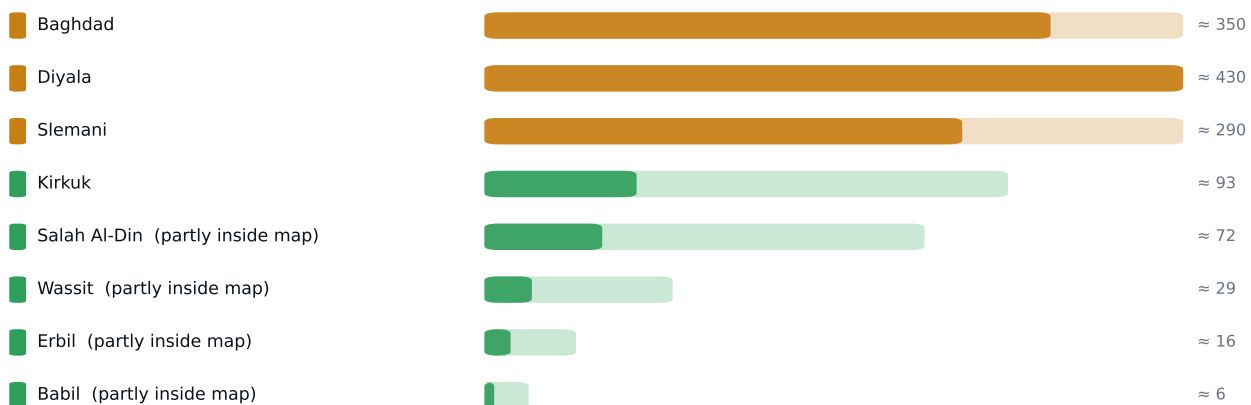


Which buildings failed

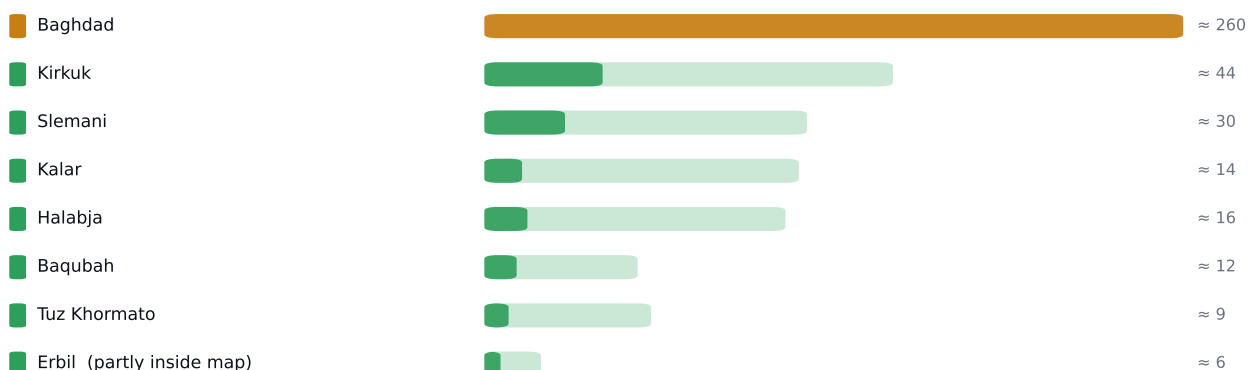
Share of each type's own buildings heavily damaged, worst first



Governorates most affected: heavily damaged buildings, central estimate and 90% range



Cities most affected



Every governorate inside the map

Area	Buildings	Heavy (P50)	90% range	% of its stock
Baghdad	2.1 million	270	140 to 1.2 thousand	0.01%
Diyala	605 thousand	250	150 to 1.8 thousand	0.04%
Slemani	668 thousand	180	100 to 1.2 thousand	0.03%
Kirkuk	419 thousand	66	33 to 320	0.02%
Salah Al-Din	226 thousand	46	23 to 270	0.02%
Wassit	124 thousand	19	9 to 120	0.02%
Erbil	221 thousand	13	6 to 56	0.01%
Babil	49 thousand	5	2 to 27	0.01%
Al-Anbar	6.3 thousand	1	none to 6	0.02%
Maysan	5	none	none to none	0%

How this report was made

Hazard engine	Bumelerze SHAKEmap 26.1.1, ground-motion models CY14, ASB14, BSSA14, KALE15, intensity via Zaniniandhofer19 (IMS-25)
Conditioning	mvn (Engler et al. 2022) on dyfi with reverse-GMICE sigma (UQ-B, D46)
Observations	1 seismic stations, 66 felt-report cells (felt channel: dyfi)
Distances	ps2ff; rupture: Origin
Damage scale	IMS-25
Provenance	config 09a6a04fa72f, product version 1, event bml20180001

Data sources

Earthquake parameters	USGS, EMSC, GEOFON
Building footprints	GFZ OpenBuildingMap, from OpenStreetMap (c) OpenStreetMap contributors, ODbL; Microsoft Building Footprints, CDLA-Permissive-2.0
Built-up surface, settlements	European Commission JRC, Global Human Settlement Layer, CC BY 4.0
Population	WorldPop, University of Southampton, CC BY 4.0
Administrative boundaries	OCHA COD-AB Iraq, CC BY-IGO; source authority Iraq Central Statistical Organisation
Building types, vulnerability	GEM Global Exposure Model; IMS-25 vulnerability table
Casualty rates (computed, not published)	FEMA Hazus Earthquake Model Technical Manual 5.1

What this is, and what it is not

An automatic estimate of shaking and building damage from the earthquake's location and magnitude, adjusted with whatever station and felt-report data existed at the time.

Cite as: Bumelerze SHAKEmap and damage report, event bml20180001, version 1, generated 21 Sep 2026, 22:30 UTC. CC BY 4.0.